

11.24 Reconsider Example 11.2 on the supply and demand for fish at the Fulton Fish Market. The data are in the file *fultonfish*.

- Add the variable *MIXED*, which indicates poor but not *STORMY* weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for $\ln(\text{PRICE})$, adding the variable *MIXED* to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of *STORMY* and *MIXED*. Is the resulting *F*-value greater than 10?
- Estimate the demand equation using *STORMY* and *MIXED* as IVs. Compare the coefficient estimates to those in Table 11.5.
- Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.
- In the reduced-form equation in part (a), test the joint significance of the indicator variables *MON*, *TUE*, *WED*, and *THU* at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

a. Call:
lm(formula = lprice ~ mon + tue + wed + thu + stormy + mixed, data = fultonfish)

Residuals:

	Min	1Q	Median	3Q	Max
	-0.92281	-0.18819	0.00813	0.22404	0.68381

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.35957	0.07915	-4.543	1.50e-05 ***
mon	-0.10825	0.10339	-1.047	0.29753
tue	-0.06609	0.10103	-0.654	0.51446
wed	-0.04927	0.10377	-0.475	0.63591
thu	0.03935	0.10071	0.391	0.69681
stormy	0.44634	0.07921	5.635	1.51e-07 ***
mixed	0.23688	0.07851	3.017	0.00321 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3413 on 104 degrees of freedom
Multiple R-squared: 0.245, Adjusted R-squared: 0.2014
F-statistic: 5.624 on 6 and 104 DF, p-value: 4.349e-05

Linear hypothesis test:
stormy = 0
mixed = 0

Model 1: restricted model
Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	106	15.876				
2	104	12.115	2	3.7604	16.14	7.857e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$P = 0.00321 < 0.05 \Rightarrow$ reject H_0 . It's significant at the 5% level.

The *F*-value is 16.14 larger than 10. *stormy* and *mixed* are strong IVs.

b. Call:
ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon + tue + wed + thu + stormy + mixed, data = fultonfish)

Residuals:

	Min	1Q	Median	3Q	Max
	-2.1327	-0.4320	0.0654	0.5133	1.1798

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.54023	0.15661	54.533	< 2e-16 ***
lprice	-0.93014	0.35340	-2.632	0.00977 **
mon	-0.01190	0.20837	-0.057	0.95456
tue	-0.52583	0.20230	-2.599	0.01068 *
wed	-0.56262	0.20696	-2.719	0.00767 **
thu	0.09987	0.20285	0.492	0.62350

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6853 on 105 degrees of freedom
Multiple R-Squared: 0.185, Adjusted R-squared: 0.1462
Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317

$M = 1 (\ln \text{price})$, IVs = *stormy*, *mixed* $\Rightarrow$ overidentified.

Table 11.5: $\ln(\text{PRICE})$ 係數為 -1.114, $sd = 0.4286$

Now: $\ln(\text{PRICE})$ 係數為 -0.93014, $sd = 0.3534 \Rightarrow$ *sd* smaller, *mixed* 為有效 IVs. #

c. Diagnostic tests:

	df1	df2	statistic	p-value
Weak instruments	2	104	16.140	7.86e-07 ***
Wu-Hausman	1	104	1.489	0.225
Sargan	1	NA	0.772	0.380

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6853 on 105 degrees of freedom
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Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317

$\frac{1}{2}$ IVs > M , H_0 : 所有 IVs 均有效.

$S = N \times R^2 = 0.772$, $P = 0.38 > 0.05$, fail to reject H_0 .

故 *mixed* 是有效 IV. #

d. Linear hypothesis test:
mon = 0
tue = 0
wed = 0
thu = 0

Model 1: restricted model
Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	108	12.408				
2	104	12.115	4	0.29256	0.6279	0.6437

$F = 0.6279$, 遠小於 10 $\Rightarrow$ fail to reject H_0 .

故星期四對 $\ln(\text{PRICE})$ 沒有足夠解釋力. 加入 *mixed* 仍無法有效 identified 供給方程. #

11.28 Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

- Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?
- Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?
- Estimate the price elasticity of demand "at the means" using the results from (b).
- Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI , PS , and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.
- What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?
- Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

a Demand: $Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di} \Rightarrow P_i = -\frac{\alpha_1}{\alpha_2} + \frac{1}{\alpha_2} Q_i - \frac{\alpha_3}{\alpha_2} PS_i - \frac{\alpha_4}{\alpha_2} DI_i - \frac{e_{di}}{\alpha_2}$

其中 $1/\alpha_2 > 0$, $-\alpha_3/\alpha_2 < 0$, $-\alpha_4/\alpha_2 < 0$.

Supply: $Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si} \Rightarrow P_i = -\frac{\beta_1}{\beta_2} + \frac{1}{\beta_2} Q_i - \frac{\beta_3}{\beta_2} PF_i + \frac{e_{si}}{\beta_2}$

其中 $1/\beta_2 > 0$, $-\beta_3/\beta_2 < 0$.

b. Call: `ivreg(formula = p ~ q + ps + di | ps + di + pf, data = truffles)`

Residuals:

	Min	1Q	Median	3Q	Max
	-39.661	-6.781	2.410	8.320	20.251

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-11.428	13.592	-0.841	0.40810
q	-2.671	1.175	-2.273	0.03154 *
ps	3.461	1.116	3.103	0.00458 **
di	13.390	2.747	4.875	4.68e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.17 on 26 degrees of freedom
Multiple R-Squared: 0.5567, Adjusted R-squared: 0.5056
Wald test: 17.37 on 3 and 26 DF, p-value: 2.137e-06

Call:

`ivreg(formula = p ~ q + pf | ps + di + pf, data = truffles)`

Residuals:

	Min	1Q	Median	3Q	Max
	-9.7983	-2.3440	-0.6281	2.4350	11.1600

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-58.7982	5.8592	-10.04	1.32e-10 ***
q	2.9367	0.2158	13.61	1.32e-13 ***
pf	2.9585	0.1560	18.97	< 2e-16 ***

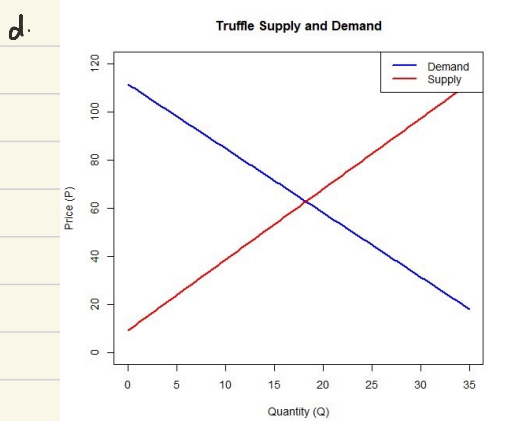
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.399 on 27 degrees of freedom
Multiple R-Squared: 0.9486, Adjusted R-squared: 0.9448
Wald test: 232.7 on 2 and 27 DF, p-value: < 2.2e-16

需求方程式中 Q 係數為 -2.671 (顯著)，符合需求法則 (P 與 Q 負相關)；供給方程式中 Q 係數為 2.937 (顯著)，符合供給法則 (P 與 Q 正相關)，所有係數均與經濟理論預期符號一致且顯著。

c. `> cat("需求價格彈性 (at the means) :", elasticity, "\n")`
需求價格彈性 (at the means) : -1.272464

在均值點估計的需求價格彈性為 -1.272，表示松露價格上漲 1%，需求量約下降 1.27%，需求具有彈性。



如圖所示，需求曲線 (藍) 向右下傾斜、供給曲線 (紅) 向右上傾斜，兩曲線交叉於均衡點附近 ($Q \approx 18$, $P \approx 63$)，符合經濟理論預期。

```
e. > cat("結構式均衡 Q*:", Q_star, "\n") > cat("簡化式均衡 P*:", P_reduced, "\n")
結構式均衡 Q*: 18.25021 簡化式均衡 P*: 62.81537
> cat("結構式均衡 P*:", P_star, "\n") > cat("簡化式均衡 Q*:", Q_reduced, "\n")
結構式均衡 P*: 62.84257 簡化式均衡 Q*: 18.2604
```

結構式求得均衡值 $Q^* = 18.25$ 、 $P^* = 62.84$ ，簡化式預測值 $Q^* = 18.26$ 、 $P^* = 62.82$ ，兩者幾乎完全一致，驗證了結構式與簡化式方法的等價性。

```
f. Call:
lm(formula = p ~ q + ps + di, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-25.0753  -2.7742  -0.4097   4.7079  17.4979

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.6195    9.0872  -1.499   0.1460
q             0.1512    0.4988   0.303   0.7642
ps            1.3607    0.5940   2.291   0.0303 *
di            12.3582    1.8254   6.770 3.48e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.814 on 26 degrees of freedom
Multiple R-squared:  0.8013,    Adjusted R-squared:  0.7784
F-statistic: 34.95 on 3 and 26 DF,  p-value: 2.842e-09
```

```
> print(comparison)
      Parameter      OLS      TSLS
(Intercept) Intercept -13.6194989 -11.428407
q              Q       0.1512043  -2.670519
ps             PS       1.3607045   3.461081
di             DI      12.3582353  13.389921
```

```
Call:
lm(formula = p ~ q + pf, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
 -8.4721  -3.3287   0.1861   2.0785  10.7513

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -52.8763    5.0238 -10.53 4.68e-11 ***
q             2.6613    0.1712  15.54 5.42e-15 ***
pf            2.9217    0.1482  19.71 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.202 on 27 degrees of freedom
Multiple R-squared:  0.9531,    Adjusted R-squared:  0.9496
F-statistic: 274.4 on 2 and 27 DF,  p-value: < 2.2e-16
```

OLS 估計的需求方程式中 Q 係數為 0.151（不顯著且符號錯誤），而 2SLS 估計為 -2.671 （顯著且符號正確），顯示 OLS 因忽略 Q 的內生性而產生嚴重偏誤，2SLS 有效修正了此問題。